

DESIGN TECHNOLOGY AND INNOVATION

Assignment 0

1. Complete the following sentence: Braille is a _____ of information designed for the visually impaired people.

☐ Vibration system

☒ Tactile system

☐ Hearing system

☐ Visual system

Yes, the answer is correct.

Score: 1

Accepted Answers: ***Tactile system***

2. What does the term SDG mean?

☐ Survival, Development and Growth

☐ Synchronic Diachronic Graphs

☐ Sustainable Design Guide

☒ Sustainable Development Goal

Yes, the answer is correct.

Score: 1

Accepted Answers: ***Sustainable Development Goal***

3. What does the word 'prosthesis' mean?

☐ An artificial eye

☐ An artificial organ

☒ An artificial device which replaces a body part

☐ An artificial nerve

Yes, the answer is correct.

Score: 1

Accepted Answers: ***An artificial device which replaces a body part***

4. The most important aspect of a helmet is?

☐ Feel

☒ Safety

☐ Shape

☐ Colour

Yes, the answer is correct.

Score: 1

Accepted Answers **Safety**

5. The surgical masks are made up of which material?

☐ Melt blown plastic

☐ Melt blown paper

☒ Melt blown fabric

☐ Melt blown nylon

Yes, the answer is correct.

Score: 1

Accepted Answers: **Melt blown fabric**

Assignment 1

1. In which of the following ways is the Jaipur foot a better alternative as opposed to the imported prosthesis in the Indian market?

☐ It was distributed to children only

☒ Because it was available at low cost

☒ It was sensitive to Indian culture, like allowing the user to sit down for praying, etc.

☒ Because it has a barefoot design

Yes, the answer is correct.

Score: 1

Accepted Answers:

Because it was available at low cost

It was sensitive to Indian culture, like allowing the user to sit down for praying, etc.

Because it has a barefoot design

2. Which of the following organizations initiated the production and distribution of the Jaipur foot across India?

☐ Sahayata Bhagwan Mahaveer Samiti

☐ Bhagwan Veer Sahayata Samiti

☒ Bhagwan Mahaveer Viklang Sahayata Samiti

☐ None of the above

Yes, the answer is correct.

Score: 1

Accepted Answers: ***Bhagwan Mahaveer Viklang Sahayata Samiti***

3. Why did D R Mehta not use the design of the prosthesis as proposed by Dr. APJ Abdul Kalam?

☐ There was a shortage of raw materials

☒ The new design was not context-based

☒ It was made using expensive materials

☐ the design was not bio friendly to human body

Yes, the answer is correct.

Score: 1

Accepted Answers:

The new design was not context-based

It was made using expensive materials

4. Among the following components of the Jaipur foot, which was getting worn out after four to five years of use?

☐ Microcellular rubber

☒ PVC shank

☐ Silicone paste

☐ Rubber socket

No, the answer is incorrect.

Score: 0

Accepted Answers: **PVC shank**

5. The Jaipur foot was designed keeping in mind which of the following disabilities?

☒ Extreme poverty

☒ Illiteracy

☒ Amputation

☐ Employment

Yes, the answer is correct.

Score: 1

Accepted Answers:

Extreme poverty

Illiteracy

Amputation

6. What are the key clusters of helmet design as discussed in the lecture 'User-centered design' by professor B. K. Chakravarthy?

☒ Ventilation cluster

☐ Design cluster

☒ Storage cluster

☐ Validation cluster

No, the answer is incorrect.

Score: 0

Accepted Answers:

Ventilation cluster

Storage cluster

7. Which layer of the helmet is mostly responsible for saving the life of an individual in a road accident?

☐ The outer shell of the helmet

☐ The PU padding of the helmet

☐ The inner shell of the helmet

☒ The thermocol layer of the helmet

Yes, the answer is correct.

Score: 1

Accepted Answers: **The thermocol layer of the helmet**

8. According to Professor B. K. Chakravarthy, why do conventional helmets fail in India?

☐ The helmet transfers shock to the brain very effectively

- ☐ They fail to prevent accidents
- ☒ The standards for the helmet are adopted from cold countries
- ☐ The helmets do not function well

Yes, the answer is correct.

Score: 1

Accepted Answers:

The standards for the helmet are adopted from cold countries

9. collapsible helmet is mainly designed as per which persona requirements?

- ☐ Kids going to school on two wheelers
- ☐ For stuntmen and mountain riders
- ☒ For middle aged office working individuals on two wheelers
- ☐ For women two wheel riders

Yes, the answer is correct.

Score: 1

Accepted Answers:

For middle aged office working individuals on two wheelers

10. According to Professor B. K. Chakravarthy, why is it important to sketch or doodle in the journey of product designing?

- ☐ For pleasure
- ☒ For externalising thought processes
- ☐ For internalizing ideas
- ☐ For better academic projects

Yes, the answer is correct.

Score: 1

Accepted Answers:

For externalising thought processes

Assignment 2

1. to the module 'Challenges of reaching a million users', which amongst the following were some common issues associated with the solar lamps in the past?

- ☐ Affordability
- ☒ Mass production
- ☒ Repairability

☒ No technical support

Yes, the answer is correct.

Score: 1

Accepted Answers:

Affordability

Repairability

No technical support

2. Which of the following options was a problem that the SoULS project has not yet managed to overcome?

☐ Local tech support

☐ After sale service

☒ Last mile tracking

☐ Making the product cheap and affordable

Yes, the answer is correct.

Score: 1

Accepted Answers:

Last mile tracking

3. As discussed by Prof. Chetan Solanki, what are the significant features of solar urja lamps?

☒ Most of the parts of the lamp are manufactured in India

☒ The lamp is easy to assemble and maintain

☐ The battery of the lamp never gets discharged

☒ The assembly of the lamp is done by local people

Yes, the answer is correct.

Score: 1

Accepted Answers:

Most of the parts of the lamp are manufactured in India

The lamp is easy to assemble and maintain

The assembly of the lamp is done by local people

4. The main objective of the SoULS project was to initially help which group of individuals?

☐ The elderly

☒ Students in the village settings

☐ Teachers of urban schools

☐ Urban teenagers

Yes, the answer is correct.

Score: 1

Accepted Answers:

Students in the village settings

5. How have the solar lamps contributed in the reduction of the number of deaths in rural settings?

☒ The lamps have helped the rural people to locate snakes in agricultural fields

- ☐ The lamps have reduced the number of suicides
- ☐ The lamps have helped the rural people for medical aid
- ☐ None of the above

Yes, the answer is correct.

Score: 1

Accepted Answers:

The lamps have helped the rural people to locate snakes in agricultural fields

6. Amongst the following Sustainable Development Goals(SDG) drafted by the United Nations, which one is associated with the SoULS project?

- ☐ Goal number 13: Climate action
- ☐ Goal number 3: Good health and well-being
- ☐ Goal number 4: Quality education
- ☒ Goal number 7: Affordable and clean energy

Yes, the answer is correct.

Score: 1

Accepted Answers:

Goal number 7: Affordable and clean energy

7. What is the main motto of the SoULS project?

- ☒ By local for local
- ☐ By landlords for landlords
- ☐ By landlords for locals
- ☐ By local for landlords

Yes, the answer is correct.

Score: 1

Accepted Answers:

By local for local

8. What step helped in the publicization of the SoUL project undertaken by Prof. Chetan Solanki?

- ☒ Through the help of government groups and panchayats
- ☒ By attempting a world record
- ☒ By reducing the profit margin so as to benefit the society
- ☐ By creating and developing a new cheaper solar panel

Partially Correct.

Score: 0.66

Accepted Answers:

Through the help of government groups and panchayats

By attempting a world record

By reducing the profit margin so as to benefit the society

9. Which of the following options form a part of the localisation process in the SoULS project?

- ☒ Employing local village women for assembling the lamp
- ☐ Drawing profit from locals to build a manufacturing plant in a metro city
- ☒ Sending representatives to teach the locals about how to repair the product
- ☒ Establishing village based charging stations managed by the locals

Partially Correct.

Score: 0.66

Accepted Answers:

Employing local village women for assembling the lamp

Sending representatives to teach the locals about how to repair the product

Establishing village based charging stations managed by the locals

10. Complete the following sentence: The independence of rural households from energy required for lighting, comfort and irrigation through self sustained solar power is called _____ ?

- ☐ Solar urja

☐ Bijli sustenance

☐ Urja sustainability

☒ Energy Swaraj

Yes, the answer is correct.

Score: 1

Accepted Answers: **Energy Swaraj**

Assignment 3

1. According to Professor Ramesh Singh, why is 'Bioengineering' the need of the hour?

☒ Because it helps to create products of future

☒ Because it helps to enhance our quality of life

☐ Because it is very sustainable

☐ Because it is very beneficial for government sectors

Yes, the answer is correct.

Score: 1

Accepted Answers:

Because it helps to create products of future

Because it helps to enhance our quality of life

2. Referring to the module 'Technology to Solution' which amongst these is NOT the main components of laser vent cleaning machines?

☐ Laser collimator and optics

☐ Compressed air jet nozzle

☐ 6 axis robotic arm

☒ Forging arm

Yes, the answer is correct.

Score: 1

Accepted Answers: ***Forging arm***

3. As discussed in the module 'Technology to Solution' by Professor Ramesh Singh, what is the main purpose of the tiny vents present inside the tyre mold?

☐ They provide better texture to the tyre

☒ They allow the gases to escape from the rubber tyre

☐ They make the molding process faster

☐ These vents are a manufacturing defect

Yes, the answer is correct.

Score: 1

Accepted Answers: ***They allow the gases to escape from the rubber tyre***

4. Professor Ramesh Singh collaborated with which of the following projects?

☐ The Jaipur foot

☒ Diamond polishing with KGK diamonds

☐ Purak - the prosthetic arm

☐ Smart Cane

Yes, the answer is correct.

Score: 1

Accepted Answers: ***Diamond polishing with KGK diamonds***

5. Referring to the module 'A Collaborative Excellence' complete the following sentence: The metal/alloy used in the TKP joint must be _____ and one such example is _____.

☐ Bio-static, titanium alloy

☐ Bio-sustainability, cobalt

☐ Bio- reactive, stainless steel

☒ Bio-compatible, titanium alloy

Yes, the answer is correct.

Score: 1

Accepted Answers: **Bio-compatible, titanium alloy**

6. Which amongst these made the total knee prosthesis (TKP) a better suited prosthesis for the local population over the imported ones?

☒ It provided flexion extension of 115 degrees

☐ It provided full axial rotation of 360 degrees

☒ It provided tibia rotation of +/-5 degrees

☐ It provided limited axial movement of the stem

Yes, the answer is correct.

Score: 1

Accepted Answers:

It provided flexion extension of 115 degrees

It provided tibia rotation of +/-5 degrees

7. The material which was used to reduce friction between moving parts of the total knee prosthesis was called?

☐ Cobalt-chromium alloy

- ☐ Titanium alloy
- ☒ Ultra high molecular weight polyethylene
- ☐ Stainless steel

Yes, the answer is correct.

Score: 1

Accepted Answers: ***Ultra high molecular weight polyethylene***

8. Which of the following is a requirement which has to be considered when designing the armamentarium and its packaging for the TKP(total knee prosthesis)?

- ☒ Ease of understanding to the doctor by providing icons and proper segmentation of the parts(On table/Off table)
- ☒ The packaging had to be weatherproof and maintain the sterilization of the tools and parts
- ☒ Spare parts should be added into the set of instruments to accommodate for worst case scenario
- ☐ The overall packaging should be aesthetically pleasing so that the set of tools sell more

Yes, the answer is correct.

Score: 1

Accepted Answers:

Ease of understanding to the doctor by providing icons and proper segmentation of the parts(On table/Off table)

The packaging had to be weatherproof and maintain the sterilization of the tools and parts

Spare parts should be added into the set of instruments to accommodate for worst case scenario

9. According to Professor B Ravi, what strategies were involved in the knee replacement process to solve the problem of variations in the length of the patient's leg?

☒ They made customised shoes with varying heel thickness to adjust minor differences in length

☒ The whole prosthesis was modular in design to accommodate for the approximate length

☐ The patient would have to go through a series of operations to correct the length of the leg

☐ The patient would have to live with the uneven leg length for the rest of his/her life

Yes, the answer is correct.

Score: 1

Accepted Answers:

They made customised shoes with varying heel thickness to adjust minor differences in length

The whole prosthesis was modular in design to accommodate for the approximate length

10. the following sentence: the tool kit which comprises the surgical instruments was commonly known by the name _____?

- ☐ TKP Kit
- ☐ Knee prosthesis kit
- ☒ Armamentarium
- ☐ Life betterment pack

Yes, the answer is correct.

Score: 1

Accepted Answers:

Armamentarium

Assignment 4

1. The core idea of the module 'Collaborative Innovation Methods' is?

- ☐ To discuss innovative ideas
- ☐ To enable organizations to start their business
- ☒ To enable organisations to generate innovative product ideas
- ☐ To discuss innovative products

Yes, the answer is correct.

Score: 1

Accepted Answers:

To enable organisations to generate innovative product ideas

2. Referring to the module 'Collaborative Innovation Methods' the design of the dual chambered bottle was inspired by which analogies?

☒ The idea of an ointment tube

☐ The idea of a cylindrical rod

☒ The analogy of the camera

☒ The idea of a traditional soda bottle

Yes, the answer is correct.

Score: 1

Accepted Answers:

The idea of an ointment tube

The analogy of the camera

The idea of a traditional soda bottle

3. Amongst the following options, which is the final pitfall in the course of innovation as discussed by Professor B K Chakravarthy?

☐ Prototyping

☐ Crafting

☒ Pilot to mass production

☐ Pilot production

Yes, the answer is correct.

Score: 1

Accepted Answers: ***Pilot to mass production***

4. Which of the 7 C's from Professor B K Chakravathy's lecture is associated with making mock ups, functional prototypes and pilot production?

☐ Check

☒ Crafting

☐ Connection

☐ Context

Yes, the answer is correct.

Score: 1

Accepted Answers: ***Crafting***

5. Why was aluminium not used as a material in the manufacturing of the Hero cycle?

☐ Because it made the cycle heavier

☐ Because it made the cycle lighter

☒ Because it was an expensive material

☐ Because it reduced the speed of the cycle

Yes, the answer is correct.

Score: 1

Accepted Answers: ***Because it was an expensive material***

6. Choose the correct sequence of pitfalls of innovation as discussed by Professor B K Chakravathy?

☐ Design - mass production - prototype

☐ Prototype - research - pilot production

☒ Prototype - pilot production - mass production

☐ Research - design action

Yes, the answer is correct.

Score: 1

Accepted Answers: ***Prototype - pilot production - mass production***

7. According to the module 'Collaborative Innovation Methods', the core team is responsible for which of the following functions?

☒ Concept detailing

☐ Tech support

☒ Concept generation and creation

☐ Manufacturing

Yes, the answer is correct.

Score: 1

Accepted Answers:

Concept detailing

Concept generation and creation

8. Which amongst these is the final C in Chakku's 7 concerns of innovation?

☐ Context - understanding the user and scenario

☐ Checklist - a clear plan of action

☐ Comprehension - arriving at design insights

☒ Connection - user testing and feedback

Yes, the answer is correct.

Score: 1

Accepted Answers: **Connection - user testing and feedback**

9. As discussed in the module 'Collaborative Innovation Methods' why was the deep drawn sheet metal concept used in the final design of the cycle?

☐ It increased the weight of the cycle

☒ It reduced the cost of the cycle by 75%

☐ It was easily available

☒ It was a faster manufacturing process

Yes, the answer is correct.

Score: 1

Accepted Answers:

It reduced the cost of the cycle by 75%

It was a faster manufacturing process

10. What is the fullform of SAPLAP analysis as discussed in the lecture 'Collaborative Innovation Methods' by Professor B K Chakravarthy?

☐ Situation actor performance learning action process

☒ Situation actor process learning action performance

☐ Situation actor performance learning ability process

☐ Situation ability performance learning action process

Yes, the answer is correct.

Score: 1

Accepted Answers: ***Situation actor process learning action performance***

Assignment 5

1) In the lecture 'Learnings from Grassroots Innovation', what did Professor Anil Gupta want to teach us from his examples?

- ☐ Design can only take place within a specialized setting
- ☒ Innovation is not necessarily dependent on educational qualification
- ☒ Real innovation comes from empathy and context
- ☒ Anyone can be innovative

Partially Correct.
Score: 0.66

Accepted Answers:

Innovation is not necessarily dependent on educational qualification

Real innovation comes from empathy and context

Anyone can be innovative

2) Referring to the module 'Learnings from Grassroots Innovation', which organisation is responsible for documenting innovation in India?

- ☐ Centre for Innovation in India (CII)
- ☒ National Innovation Foundation
- ☐ Honey Bee Network (HBN)

☒ Indian Innovation Organisation (IIO)

Yes, the answer is correct.
Score: 1

Accepted Answers: *National Innovation Foundation*

3) Which example suits the definition of autopoiesis the best amongst the following, as discussed in the module 'Learnings from Grassroots Innovation'?

☒ A knife becoming sharper as it cuts

☐ A knife getting blunt after use

☐ A walking cane that vibrates

☐ A self recharging mobile phone

Yes, the answer is correct.
Score: 1

Accepted Answers: *A knife becoming sharper as it cuts*

4) Professor Anil Gupta undertakes which of the following tours with his students to show them the various unique innovations across the rural landscape?

☒ Shodh yatra

☐ Rath Yatra

☐ Dandi March

☐ Van yatra

Yes, the answer is correct.
Score: 1

Accepted Answers:

Shodh yatra

5) What is the main purpose of Shodh Yatra?

☒ To experience the rural lifestyle in extreme climate conditions

☐ To experience the rural lifestyle during the harvest season

☐ To reach the top of shodha Devi temple

☒ To learn from nature

Yes, the answer is correct.
Score: 1

Accepted Answers:

To experience the rural lifestyle in extreme climate conditions

To learn from nature

6) As per the module 'Learnings from Grassroots Innovation', what are the three V's as discussed by Professor Anil Gupta?

☒ Voice

☒ Velocity

☒ Visibility

☐ Vice

Yes, the answer is correct.
Score: 1

Accepted Answers:

Voice

Velocity

Visibility

7) As per the lecture 'Learnings from Grassroots Innovation', the main objectives of the Honey bee network are?

☒ Looks for creative people

☒ Celebrate creative innovations

☒ Sense unmet social needs

☐ Localization

Yes, the answer is correct.
Score: 1

Accepted Answers:

Looks for creative people

Celebrate creative innovations

Sense unmet social needs

8) What message does Professor Anil Gupta teach us through the examples used in the lecture 'Learnings from Grassroots Innovation'?

☐ Design can only happen within a specialized context

☐ Innovation is dependent on educational qualifications

☒ Real innovation comes from empathy and context

☒ Minimum material maximum efficiency

Yes, the answer is correct.
Score: 1

Accepted Answers:

Real innovation comes from empathy and context

Minimum material maximum efficiency

9) According to Professor Anil Gupta, sectors, seasons, stage, social segments and structures of governance are?

- ☐ The 6 features of innovation
- ☐ The 6 characteristics of development
- ☐ The 6 pillars of sustainability
- ☒ The 6 dimensions of inclusion/exclusion

Yes, the answer is correct.
Score: 1

Accepted Answers: *The 6 dimensions of inclusion/exclusion*

10) As discussed in the module 'Learnings from Grassroots Innovation' what are the three important branches of Innovation Incubation Network?

- ☐ Material, financial, technology
- ☒ Innovation, enterprise, investments
- ☐ Innovation, technology, education
- ☐ Culture, society, innovation

Yes, the answer is correct.
Score: 1

Accepted Answers:

Innovation, enterprise, investments

Assignment 6

1) As discussed in the lecture 'Systematic Approach to Biomed Innovation', what does BETiC stand for?

☒ Biomedical Engineering & Technology Incubation Centre

☐ Biomedical Engineering & Technical Innovation Centre

☐ Biomedical Engineering & Technology Innovation Centre

☐ Biomedical Engineering & Technical Incubation Centre

Yes, the answer is correct.
Score: 0

Accepted Answers:

Biomedical Engineering & Technology Incubation Centre

2) What are the filters that should be kept in mind while designing a biomedical device as discussed by Professor B Ravi?

☒ Significant market requirement

☒ Capability and feasibility

☒ Value proposition

☐ Detailed design

Yes, the answer is correct.
Score: 1

Accepted Answers:

Significant market requirement

Capability and feasibility

Value proposition

3) According to the module on Systematic Approach to Biomed Innovation, which step does NOT come under the stage of 'Deploy'?

☐ Business model

☐ Funding

☐ Intellectual property rights

☒ Concept and feasibility of the product

Yes, the answer is correct.
Score: 1

Accepted Answers:

Concept and feasibility of the product

4) Which of the following qualities is the innovative factor/s about the Orthopaedic Hybrid Plaster Splint?

☒ It has the ability to custom fit the user

☐ It came in a kit with many parts

☒ It is easy to apply

☐ All of the above

Yes, the answer is correct.

Score: 1

Accepted Answers:

It has the ability to custom fit the user

It is easy to apply

5) Referring to the module Systematic Approach to Biomed Innovations, which of the following are the essential criteria for patentability?

☒ Novel - not publicly known

☒ Non-obvious to others

☐ Intellectual property

☒ Useful - practical application

Yes, the answer is correct.
Score: 1

Accepted Answers:

Novel - not publicly known

Non-obvious to others

Useful - practical application

6) According to Professor B Ravi, how can designers protect their designs and products from other competitors?

☐ Through financial management

☐ Through human resource management

☒ Through intellectual property management

☐ Through product quality management

Yes, the answer is correct.
Score: 1

Accepted Answers:

Through intellectual property management

7) As per the module 'Systemic Approach to Biomed Innovations' why is it necessary to promote novel medical devices in India?

- ☒ Because the existing medical devices are expensive and imported
- ☒ Because there is an increasing need for affordable medical devices
- ☐ Because the existing medical devices are unsafe
- ☐ Because we have enough resources to do so

Yes, the answer is correct.
Score: 1

Accepted Answers:

Because the existing medical devices are expensive and imported

Because there is an increasing need for affordable medical devices

8) According to Professor B Ravi, why is there a need to focus upon sustainable development goal number 3 (Good health and well being)?

- ☒ Health care is universal need
- ☒ Low cost of medical devices can have significant social impact
- ☐ Health care sector is very profitable for designers

☒ Funding avenues and incubators are available today

Yes, the answer is correct.
Score: 1

Accepted Answers:

Health care is universal need

Low cost of medical devices can have significant social impact

Funding avenues and incubators are available today

9) Who amongst these are the important stakeholders according to Professor B Ravi as discussed in the lecture 'Systemic approach to Biomed Innovations'?

☒ Doctors

☒ Entrepreneurs

☐ Manufacturers

☒ Researchers

Yes, the answer is correct.
Score: 1

Accepted Answers:

Doctors

Entrepreneurs

Researchers

10) Choose the correct sequence of the stages involved in the process of medical device innovation as discussed in the module 'Systemic Approach to Biomed Innovations'?



Define, Deploy, Develop, Deliver



Define, Develop, Deliver, Deploy



Define, Deliver, Deploy, Develop



Develop, Define, Deploy, Deliver

Yes, the answer is correct.
Score: 1

Accepted Answers:

Define, Develop, Deliver, Deploy

Assignment 7

1) What are the essential elements of Task Classification as mentioned in the lecture by Professor Amaresh Chakrabarti?



Society



Technology



Business

☐ Development

Yes, the answer is correct.
Score: 1

Accepted Answers:

Society

Technology

Business

2) Complete the following sentence: A solution and a product that functions well in different situations and by different users is a _____.

☒ Robust design

☐ Sturdy design

☐ Innovative design

☐ Flexible design

Yes, the answer is correct.
Score: 1

Accepted Answers:

Robust design

3) Complete the following sentence: It is a win-win for society and business if _____

☒ Both profit and value are acceptable

☐ If only profit is acceptable and not value

☐ If only value is accepted and not profit

☐ None of the above.

Yes, the answer is correct.
Score: 1

Accepted Answers:

Both profit and value are acceptable

4) How did Sickle Innovations make a comeback from its initial failure after its cotton picking machines?

☒ The company made other alternative versions of farming tools/machines

☐ The company used social media promotions about its success

☒ The company reanalysed the market and made products where it was really needed

☒ The company converted from a single product company to a franchise

Partially Correct.
Score: 0.67

Accepted Answers:

The company made other alternative versions of farming tools/machines

The company reanalysed the market and made products where it was really needed

The company converted from a single product company to a franchise

5) Which of the following is a condition for a successful product according to Professor Amaresh Chakrabarti?

- ☐ The product should be known by a large group of people
- ☐ A large group of people must work on the development of the product
- ☒ All members in the chain of supply of the product must be benefitted
- ☐ Extensive marketing of the product is key

Yes, the answer is correct.

Score: 1

Accepted Answers:

All members in the chain of supply of the product must be benefitted

6) A designer can manipulate which of the following aspects according to the module 'Research to Innovation'?

- ☐ Value
- ☒ Performance

☐ Manufacturing process

☒ Cost

No, the answer is incorrect.
Score: 0

Accepted Answers:

Performance

Cost

7) According to the lecture 'Research to Innovation' what kind of innovation was used in the Rolls Royce Trent turbine engine to stop the formation of ice at the cone?

☐ The cone was heated at a higher temperature

☐ The cone was attached with the wiggle motor

☐ The cone was sharpened

☒ The cone was made with flexible material

Yes, the answer is correct.
Score: 1

Accepted Answers:

The cone was made with flexible material

8) What were some of the major challenges involved in Sickle Innovations in early stages as discussed by Professor Amaresh Chakrabarti?

- ☒ Use of plastic parts were investment heavy
- ☒ Few parts wore out faster in field conditions
- ☒ Morning dew on cotton lint challenged the overall performance of the machine
- ☐ Use of aluminium alloy was very expensive

Yes, the answer is correct.
Score: 1

Accepted Answers:

Use of plastic parts were investment heavy

Few parts wore out faster in field conditions

Morning dew on cotton lint challenged the overall performance of the machine

9) According to Professor Amaresh Chakrabarti, why did the cotton picking machine fail to reach its end users?

- ☐ Because the cotton was going to ginning mills
- ☒ Because it ended up empowering the company that is selling cotton
- ☐ Because the middle men were losing profit from it

☐ Because it was not approved by the government

Yes, the answer is correct.
Score: 1

Accepted Answers:

Because it ended up empowering the company that is selling cotton

10) As discussed in the module Research to Innovation, which of the following knowledge systems are essential for the successful execution of an innovation?

☒ Domain knowledge

☐ Intellectual knowledge

☐ Cultural knowledge

☒ Process knowledge

Yes, the answer is correct.
Score: 1

Accepted Answers:

Domain knowledge

Process knowledge

Assignment 8

1) As discussed in the module 'Smart Cane for the Blind', why was audio technology not used as a feedback mechanism for the users?

☒ Because the visually impaired people were unable to hear audio alerts in loud locations

☒ Because the visually impaired people did not want to declare their blindness to others

☐ Because it was an expensive technological component

☐ Because it was confusing for the user

Yes, the answer is correct.

Score: 1

Accepted Answers:

Because the visually impaired people were unable to hear audio alerts in loud locations

Because the visually impaired people did not want to declare their blindness to others

2) As mentioned in the lecture 'Smart Cane for the Blind', the combination of which elements is necessary for an innovation to happen?

☒ Design

☒ Policy

☒ Technology

☐ Environment

Yes, the answer is correct.

Score: 1

Accepted Answers:

Design

Policy

Technology

3) According to the module 'Smart Cane for the Blind', why did standardized english braille labelling embossed on the smartcane fail as a design feature?

☒ Because the users read and understood braille in different languages

☐ So as to provide a finishing look to the smartcane

☐ Because adding a braille would increase the manufacturing costs of the cane

☐ It would detect obstacles better

Yes, the answer is correct.

Score: 1

Accepted Answers:

Because the users read and understood braille in different languages

4) Which of the following were some of the unique features of the Smart Cane as discussed in the module 'Smartcane for the Blind' by Professor Madhusudhan Rao?

☒ It is splash proof

☒ It is attachable to existing white cane

☐ It is unbreakable

☐ It is automatic

Yes, the answer is correct.
Score: 1

Accepted Answers:

It is splash proof

It is attachable to existing white cane

5) According to Professor Madhusudhan Rao, what important learnings were acquired by interacting with visually impaired people?

☒ The users needed empowerment rather than sympathy

☐ The users needed financial help

☐ The users needed employment

☒ The users wanted to integrate themselves with the rest of the society

Yes, the answer is correct.
Score: 1

Accepted Answers:

The users needed empowerment rather than sympathy

The users wanted to integrate themselves with the rest of the society

6) Referring to the lecture 'Smartcane for the blind', why do whitecanes fail to help the visually impaired in preventing collisions?

- ☐ Because it is very heavy
- ☒ Because it is designed to only sense obstacles knee below
- ☐ Because it does not work in open spaces
- ☐ Because it is difficult to manage

Yes, the answer is correct.
Score: 1

Accepted Answers:

Because it is designed to only sense obstacles knee below

7) Referring to the lecture 'Smartcane for the blind', why did Professor Madhusudhan Rao find it important to study the ergonomics of gripping?

- ☒ Because it was essential to understand the pressure points of the hand
- ☒ Because the users complained about the pain caused by continuous holding of the smartcane
- ☐ Because it would help to understand the relation between human factors and ergonomics
- ☐ Because it would make it an innovation

Yes, the answer is correct.
Score: 1

Accepted Answers:

Because it was essential to understand the pressure points of the hand

Because the users complained about the pain caused by continuous holding of the smartcane

8) For the visually impaired, what is the most fundamental aspect amongst these as discussed in the module 'Smartcane for the blind'?

☒ Mobility

☐ Safety

☐ Employment

☐ Security

Yes, the answer is correct.
Score: 1

Accepted Answers: *Mobility*

9) As per the module 'Smartcane for the blind' why was the design of 5 folds incorporated into the smartcane instead of 4 folds?

☐ To make it look stylish

☒ To make it space saving for women to keep it in purse

☐ To make it longer in length

☐ To make it easy to carry

Yes, the answer is correct.
Score: 1

Accepted Answers:

To make it space saving for women to keep it in purse

10) Which of the following materials were used in the making of the smart cane?

☒ Biocompatible plastic

☐ Rubber coating

☐ Titanium linings

☐ Clay

Yes, the answer is correct.
Score: 1

Accepted Answers:

Biocompatible plastic

